

AYUSHMAN SINGH

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Technical Experience

Parakh AI – AI Interviewer | [Link](#)

Nov 25 - present

- **Full-Stack Development:** Built an end-to-end platform using FastAPI and React.js with PostgreSQL, facilitating seamless candidate and company interactions.
- **AI Interviewer:** Integrated Google Gemini Pro with STT/TTS to conduct dynamic, voice-enabled technical interviews with automated candidate evaluation.
- **Automated Shortlisting:** Developed an AI engine to parse resumes (PyPDF2) and rank candidates based on job description semantic matching.
- **Computer Vision Proctoring:** Implemented real-time face and eye tracking using OpenCV to monitor interview integrity and detect malpractice.
- **Secure Auth & Messaging** Designed JWT and OAuth 2.0 security layers with OTP-based email verification via Resend.
- **Deployment:** Containerized the application with Docker and deployed the optimized frontend via Vercel and Vite.
- **Tech Stack:** FastAPI, React, PostgreSQL, Gemini AI, OpenCV, Docker, JWT.

Brain AI Multi-Agent Coding Assistant | [Link](#)

March 26- present

Architected a Role-Based Multi-Agent system (Architect, Coder, Tester, DevOps) to orchestrate the end-to-end SDLC, capable of generating, testing, and packaging complex desktop applications (e.g., a full code editor) into deployable executables in under 15 minutes.

- Implemented an embedded H2 Database for persistent storage of project history, user settings, and dynamic role-specific system prompts.
- Developed a JavaFX desktop application integrating Google's Gemini 2.5 Flash for high-performance AI coding assistance.
- Built an API Key Management system with auto-rotation and failover logic, ensuring 99% uptime by handling rate limits and invalid keys.
- Engineered a multi-threaded architecture to decouple UI logic from AI network calls, significantly enhancing application responsiveness.
- Streamlined cross-platform build pipelines using PowerShell and Bash to generate native Windows (.msi) installers and modular binaries.
- **Multi-Agent Orchestration:** Chained specialized agents for automated planning, coding, and QA.
- **Database Engineering:** Managed state and configuration using SQL (H2) for offline-first reliability.
- **AI Resilience:** Engineered Exponential Backoff and programmed key failover to mitigate API instability.
- **DevOps:** Standardized distribution with native installers and custom build scripts.
- **Concepts:** Multi-threading, API Key Rotation, Role-Based Orchestration, Exponential Backoff, SDLC Automation.
- **Tech Stack:** JAVA FX , JDBC , Google Gemini 2.5 Flash, Oracle xe.

Skills

Languages & Core Tech: Java (JavaFX, JDBC, Multi-threading), Python (FastAPI), JavaScript (React.js, Vite), SQL, Bash, PowerShell

Artificial Intelligence: Multi-Agent Orchestration, LLM Integration (Google Gemini), Computer Vision (OpenCV), NLP/Parsing (PyPDF2), STT/TTS

Databases & DevOps: Oracle XE, , Git, Docker, Vercel, Render

Architecture & Auth: JWT, OAuth 2.0, API Key Failover/Rotation, Exponential Backoff, Role-Based System Architecture

Developer & AI Tools: Google Antigravity, Cursor, Gemini CLI, Neovim, Postman, Google Colab

Professional Skills: Effective Technical Communication, Cross-Functional Team Collaboration, Project Pitching, and Strategic Problem Solving

Education

Master of Computer Applications (MCA) Sept 2025 – Present | KIET Deemed to be University | Delhi NCR, India | Relevant Coursework: Web Development, Agentic AI Development, Advanced Java

Bachelor of Computer Applications (BCA) Sept 2022 – Aug 2025 | Deen Dayal Upadhyaya Gorakhpur University | Gorakhpur, India | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (SQL), C++

Achievements

- **Top 5% Globally (Top 2,000 / 30,000+):** GDG Devpost Challenge for the "Parakh AI" platform.
- **Winner:** Google Developer Groups (GDG) Campus Competition.
- **2nd Runner-Up:** Innotech (Branch Division).
- **Finalist:** Advanced to the final round of the HackoClock Hackathon.